An evaluated complete C1 certificate when the angular coefficient is nonpositive

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This note treats only the $\lambda \le 0$ branch of `Uniform_C1_Certificate.tex`, on the source and independent testing rectangle of `fixed_box_c1.tex`. It supplies a complete parent certificate, including the retained positive squares. It is not a global lift or a certificate for other testing charts.

1 Domain and source bounds

Use exactly the rational source formulas and strict gaps of `fixed_box_c1.tex`, except that $\lambda \le 0$. Thus

$$0 < x, \sigma \le \varepsilon = 2^{-160}, \quad 0 < z \le 1, \quad 0 < b < 1, \quad r = bz < 1, \quad 0 < h < 1, \quad D, \beta \ge 0,$$
$$0 \le u, w \le \delta = 2^{-100}.$$

State coordinates W, X, Y, Z are unrestricted. Set $k = \sqrt{x}$, $q = \sqrt{z}$, $Q = 1-b$, $H = h/\sigma$, $H_0 = 1+zQ$, $d_G = 4D/\sigma^2$, $L = \lambda/\sigma$, and $K_c = 1 + L^2$. The source identities BE3–BE4 give the following finite bounds:

$$H \le 2, \quad h \le 2\sigma, \quad -8 \le L \le 0, \quad |d_G - 2H_0| \le 200\varepsilon, \quad |4\kappa H - 2H_0| \le 400\varepsilon. \tag{N1}$$

The already established common bounds used below are

$$1 \le H_0 \le 2, \quad |\nu| \le 3, \quad p \le 2, \quad m \le 6\varepsilon^2, \quad D \le 4\sigma^2, \quad |R_0| \le 512\varepsilon, \quad |B_x| \le 48\varepsilon,$$
$$|R_x|, |A_y| \le 32, \quad |\chi| \le 64, \quad |k_1| \le 16, \qquad 4a, 4\ell \ge 1. \tag{N2}$$

Here all coefficient symbols have their displayed definitions in the "finite formula for the main Gram" section of `fixed_box_c1.tex`, and $R_\gamma = R_0 - 8pu^2$. The coefficient bounds for E_{yy}, E_{RR}, E_{yR} on their respective testing supports (u^2, w^2) and $(1, u^2, w^2)$ are

$$(4, 4), \qquad (64\sigma, 16, 4), \qquad (32\sigma, 12, 8). \tag{N3}$$

These bounds retain $C(1 - r)$ rather than estimating the possibly unbounded positive coefficient C. Their source-formula proofs in the shared calculation do not use $\lambda > 0$; only the subsequent positive-angular construction there uses that sign.

Here is a short derivation of (N1), using only the shared source formulas and common bounds. Multiplying $\lambda \le 0$ by σ gives

$$8Dh + 4\beta\theta^2 + 2h^2 \le 4bp\sigma^2 z.$$

Thus $H^2 \le 2bpz \le 4$, $h \le 2\sigma$, and $\beta\theta^2/\sigma^2 \le bpz \le 2$. In particular $\theta \ge 1 - 2\varepsilon$ and $\beta\theta/\sigma^2 \le 3$. Also every positive term in L can be discarded to give $L \ge -4bpz \ge -8$. For the remaining estimates it

is enough to use $\varepsilon \le 1/2000$. The positive formula for P, with $h \le 2\sigma$, yields $1 \le P \le 1+4\varepsilon$, and $1 \le B_s \le 1+6\varepsilon$. BE3 is exactly

$$HB_s - H_0 = 2P\beta\theta/\sigma.$$

Consequently

$$|H - H_0| \le \left\{ \frac{4(1+4\varepsilon)}{1-2\varepsilon} + 12 \right\} \varepsilon < 18\varepsilon.$$

Use the common bound $|\mathcal{H} - H_0| \le 32\varepsilon$ and $|PJ - 1| \le 6\varepsilon$, $PJ \ge 1-\varepsilon$. BE4 gives

$$d_G = \frac{2\mathcal{H}}{PJ} - \frac{4x\beta\theta}{J\sigma^2}.$$

The first term differs from $2H_0$ by at most $88\varepsilon/(1-\varepsilon)$ and the second has absolute value at most $12\varepsilon/(1-\varepsilon)$, so their sum is less than 200ε. Finally $|M-1| \le \varepsilon$ implies $|MP-1| \le 6\varepsilon$, while $1 \le K(1+\sigma r)(1+\sigma z) \le 1+4\varepsilon$. The formula for κ therefore gives $|\kappa - 1/2| \le 5\varepsilon/(1-\varepsilon) < 39\varepsilon$. Together with $H \le 2$, this proves

$$|4\kappa H - 2H_0| \le 8(39\varepsilon) + 2(18\varepsilon) < 400\varepsilon.$$

2 Exact complete quadratic correction

Retain the uniform proof's roots

$$N_x = X + uhZ, \quad N_y = (Y - bwZ)/q, \quad R = wZ/q - uqW, \quad \Gamma = \sigma Z - uwW,$$
$$A_\theta = q(u^2 - x)W, \quad V = uw(u^2 - x)W, \quad J_C = N_y + bR, \quad J_\ell = N_y - QR, \quad X_e = X + ewW.$$

In particular $R = J_C - J_\ell$ and $N_y = QJ_C + bJ_\ell$. Introduce the source-only shear

$$T = \Gamma + (L/2)V, \qquad g = (N_x, J_C, J_\ell, T, V, A_\theta)^{\mathsf{T}}, \qquad D_* = \operatorname{diag}(2,1,1,1,1,2).$$

This shear is invertible with inverse $\Gamma = T - (L/2)V$; both directions have coefficients bounded by four. The source baseline has the exact decomposition

$$\begin{aligned} 2H_0\Gamma^2 + L\Gamma V + (1+L^2)V^2 &= T^2 + V^2 + (2H_0 - 1)\Gamma^2 + \tfrac{3}{4}L^2V^2, \\ 2H_0A_\theta^2 &= 2A_\theta^2 + 2(H_0 - 1)A_\theta^2. \end{aligned} \tag{N4}$$

Thus the principal baseline is the fixed diagonal D_*, with the last terms in (N4) retained as nonnegative source squares. This is a source-only change of roots, not a congruence depending on testing parameters. Below all final directions are fixed in these six coordinates.

Define $\mathcal{E}$ by the following quadratic polynomial, substituting $\Gamma = T - (L/2)V$, $R = J_C - J_\ell$, and $N_y = QJ_C + bJ_\ell$:

$$\begin{aligned} \mathcal{E} = {} & E_{yy}N_y^2 + E_{RR}R^2 + E_{yR}N_yR + (d_G - 2H_0)\Gamma^2 - 4\nu uN_x\Gamma + qw(R_0 - 8pu^2)J_C\Gamma \\ & - 4uqw(2m + 2D + 1)N_xN_y + uqwR_xN_xR + qwB_xN_xA_\theta + uA_yN_yA_\theta \\ & + (\chi + \lambda)uA_\theta R + (4\kappa H - 2H_0)A_\theta^2 + Lu^2\Gamma^2 \\ & - K_c\sigma u^2A_\theta^2 - K_c\sigma u(u^2 - x)A_\theta R + K_c(u^2 - x)V\Gamma. \end{aligned} \tag{N5}$$

This specifies every entry of G by $g^{\mathsf{T}}Gg = g^{\mathsf{T}}D_*g + \mathcal{E}$; cross coefficients are divided by two between symmetric entries.

For completeness the entire retained part is

$$\begin{aligned}\Pi &= 4\beta N_x^2 + (4\beta + 2 + 4m)(wN_x)^2 + 2(2D+1)(uX_e)^2 + 4bpz(u\Gamma)^2 \\ &\quad + 4a\{(uJ_C)^2 + (wJ_C)^2 + (uwJ_C)^2\} + 4\ell(uJ_\ell)^2 + 4Cz(wJ_C)^2 \\ &\quad + [4a + 2z(2C+1) - 1]J_C^2 + (4\ell - 1)J_\ell^2 \\ &\quad + (2H_0 - 1)\Gamma^2 + \tfrac{3}{4}L^2V^2 + 2(H_0 - 1)A_\theta^2.\end{aligned} \tag{N6}$$

Every coefficient in (N6) is nonnegative in the source field. No mixed radical occurs.

The exact parent identity is

$$F_w = \Pi + g^{\mathsf{T}} G g. \tag{N7}$$

Indeed the uniform proof gives

$$K_A A_\theta^2 + \mathscr{S} = 4\kappa H A_\theta^2 + Lu^2\Gamma^2 + L\Gamma V + \lambda u A_\theta R.$$

Add K_c times its exact zero identity

$$V^2 - \sigma u^2 A_\theta^2 - u\sigma(u^2 - x)A_\theta R + (u^2 - x)V\Gamma = 0.$$

Splitting the radial and N_x pivots and then using (N4) gives precisely (N5)–(N7), including both complete Gamma normal terms.

3 Evaluated coefficient bound and fixed directions

All entries of $G - D_*$ are polynomials on the fixed testing support

$$\{1, w, w^2, u, uw, u^2, u^2w, u^3\}. \tag{N8}$$

Here is a positive-coefficient majorant for the absolute coefficient polynomial of (N5). Treat its root letters as formal nonnegative variables, put $n = J_C + J_\ell$, and $C_* = T + 4V$. Then

$$\begin{aligned}\mathcal{Q} &= (4u^2 + 4w^2)n^2 + (64\varepsilon + 16u^2 + 4w^2)n^2 + (32\varepsilon + 12u^2 + 8w^2)n^2 \\ &\quad + 200\varepsilon C_*^2 + 12uN_xC_* + (512\varepsilon w + 16u^2w)J_CC_* \\ &\quad + 16uwN_xn + 32uwN_xn + 48\varepsilon wN_xA_\theta + 32unA_\theta + 65uA_\theta n \\ &\quad + 400\varepsilon A_\theta^2 + 8u^2C_*^2 + 65\varepsilon u^2A_\theta^2 + 65\varepsilon(u^3 + u)A_\theta n + 65(u^2 + \varepsilon)VC_*.\end{aligned} \tag{N9}$$

For the two replacement terms use $|\lambda| \le 8\varepsilon$, $|\chi + \lambda| \le 65$, $K_c \le 65$, and $x, \sigma \le \varepsilon$. All other terms use (N1)–(N3). In particular $|2m + 2D + 1| \le 4$, a deliberately loose bound.

Evaluating this positive polynomial at one gives total coefficient norm $10189 < 2^{14}$. Its constant-testing part has norm 6109ε. The successive summands contribute

$$32, 336, 208, 5000, 60, 2640, 32, 64, 48, 64, 130, 400, 200, 65, 260, 650$$

to the first total. The constant part consists of $(256 + 128 + 5000 + 400 + 325)\varepsilon$. Consequently the deliberately larger rational bounds

$$|G_{ij}(0,0) - (D_*)_{ij}| \le 2^{20}\varepsilon, \quad \text{every nonconstant coefficient has magnitude } \le 2^{20} \tag{N10}$$

hold in the source field.

For six coordinates use the exact fixed-direction decomposition

$$G = \sum_i (G_{ii} - 5/16) e_i e_i^{\mathsf{T}} + \sum_{i<j} \frac{1/16 + G_{ij}}{2} (e_i + e_j)(e_i + e_j)^{\mathsf{T}} + \sum_{i<j} \frac{1/16 - G_{ij}}{2} (e_i - e_j)(e_i - e_j)^{\mathsf{T}}. \tag{N11}$$

There are 36 scalar channels. Let $b_1, \dots, b_7$ be the nonconstant monomials of (N8), fix

$$c_0 = 2^{-8}, \qquad A_* = 2^{30}, \qquad \gamma = \frac{1}{7\,2^{38}},$$

and write a channel as $q_0(c) + \sum_j a_j(c) b_j(t)$. The diagonal constant channels are at least $11/16 - 2^{20}\varepsilon$; the off-diagonal ones are at least $1/32 - 2^{19}\varepsilon$. Thus $q_0 \geq c_0$ and $|a_j| \leq A_*$. Also $0 \leq b_j \leq \delta < \gamma$ throughout the independent testing box. The exact separated identity

$$q_0 + \sum_j a_j b_j = (q_0 - c_0) + \frac{1}{2} \sum_{j=1}^{7} \{(A_* + a_j)(\gamma + b_j) + (A_* - a_j)(\gamma - b_j)\} \tag{N12}$$

uses $7A_*\gamma = c_0$. Every factor on its right is nonnegative in its own labelled field. Multiplying these squares by the fixed linear directions in (N11) proves (N7) by an evaluated finite tensor sum of squares, without an infinitesimal-residue argument on a finite rectangle.

4 A fixed complete testing dictionary

Every coordinate of g is source-linearly spanned by

$$\mathcal{V}_0 = \{X, Y, Z, W, uZ, wZ, uW, uwW, u^2W, u^3wW\}. \tag{N13}$$

For example $J_C = Y/q - buqW$, $J_\ell = Y/q - wZ/q + QuqW$, and $T = \sigma Z - uwW + (L/2)(u^3wW - xuwW)$. Thus the main Gram uses the 150 displayed functions

$$\mathcal{V}_0 \;\cup\; \{f\sqrt{\gamma + b_j},\; f\sqrt{\gamma - b_j} : f \in \mathcal{V}_0,\; 1 \leq j \leq 7\}. \tag{N14}$$

All retained roots in (N6) lie in the source-linear span of $\mathcal{V}_0$ together with the seven additional functions

$$\{wX, uwZ, uX, u^2wW, uY, wY, uwY\}. \tag{N15}$$

For instance $uX_e = uX + euwW$ and $u\Gamma = \sigma uZ - u^2wW$. This proves a complete-parent dictionary of 157 displayed testing functions. Each source-only linear combination obtained from (N11)–(N12) and (N6) has a nonnegative source coefficient, absorbed by its own source square root. The formulas give the finite coefficient prescription; no successful global pencil or global feature bound follows.

For signed testing angles compose the entire dictionary with the source-independent gauge of `fixed_box_c1.tex`; this does not change the count. The polynomial identities specialize literally at $u = 0$, $w = 0$, and arbitrary zero state coordinates. Applying the same real certificate to the real and imaginary parts treats complex states (with the corresponding doubled real dictionary).

No inverse of D or β appears. The $\beta = 0$ face is allowed literally. In this branch (N1) actually implies $d_G \geq 2 - 200\varepsilon > 0$, so the $D = 0$ face has empty intersection with the stated box and $\lambda \leq 0$. Actual $D = 0$ sources are therefore supplied by the positive-λ branch, not by this note. This distinction is a consequence of the explicit bounds, not an additional exclusion.

Nonempty sources, including the zero-beta face

For a completely explicit source path set

$$x = \sigma = t, \qquad z = b = 1 - t^2, \qquad h = t - 4t^2 + Bt^3, \qquad 16 \le B \le 18, \quad 0 < t \le \varepsilon.$$

All strict native gaps hold, since $0 < h \le 2t$, $E = 1 - h > 0$, and $J, M \ge 1 - t > 0$. Let $p_N = 2Pp$, $D_N = 2JPD$, $\beta_N = 2\theta P\beta$, and

$$N_L = 4D_N h + 2\beta_N \theta J + 2h^2 JP - 2bp_N \sigma^2 zJ = \sigma^2 JPL.$$

Direct polynomial multiplication, grouping by powers of t, gives

$$\begin{aligned} |D_N - t^2| &\le 27640t^3, \\ |\beta_N - (B-17)t^3| &\le 24459t^4, \\ |N_L - (2B-38)t^3| &\le 248856t^4. \end{aligned}$$

For these bounds take absolute coefficient sums in (t, B) and replace every B^j by 18^j in the remainders. Thus $D > 0$ and $L < 0$ throughout this path. At $B = 18$, also $\beta > 0$, giving an entirely rational strict source for each rational t. The signs of β_N at $B = 16, 18$ are opposite. Polynomial continuity therefore gives $B \in (16, 18)$ with $\beta = 0$, while $D > 0$ and $L < 0$ remain strict. This verifies actual zero-beta sources in the branch; it is not only a formal boundary allowance.